Implementation in Python:

```
from sklearn.metrics import accuracy_score

# Example usage
accuracy = accuracy_score(y_test, predictions)
print("Accuracy:", accuracy)
```

Limitations:

- Accuracy can be misleading for imbalanced datasets where one class dominates. For example, if 95% of instances belong to one class, a model that always predicts that class will have high accuracy but poor performance on the minority class.

2. Precision and Recall

Precision and recall are particularly important when dealing with imbalanced datasets, where the costs of false positives and false negatives differ significantly.

Precision

Precision measures the proportion of correctly predicted positive instances out of all predicted positives. It answers the question: "Of all the instances predicted as positive, how many are truly positive?"

Formula:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall

Recall (also known as sensitivity) measures the proportion of correctly predicted positive instances out of all actual positives. It answers the question: "Of all the true positive instances, how many did the model correctly identify?"

Formula:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Implementation in Python:

```
from sklearn.metrics import precision_score, recall_score # Example usage
precision = precision_score(y_test, predictions) recall = recall_score(y_test, predictions)
print("Precision:", precision)
print("Recall:", recall)
```